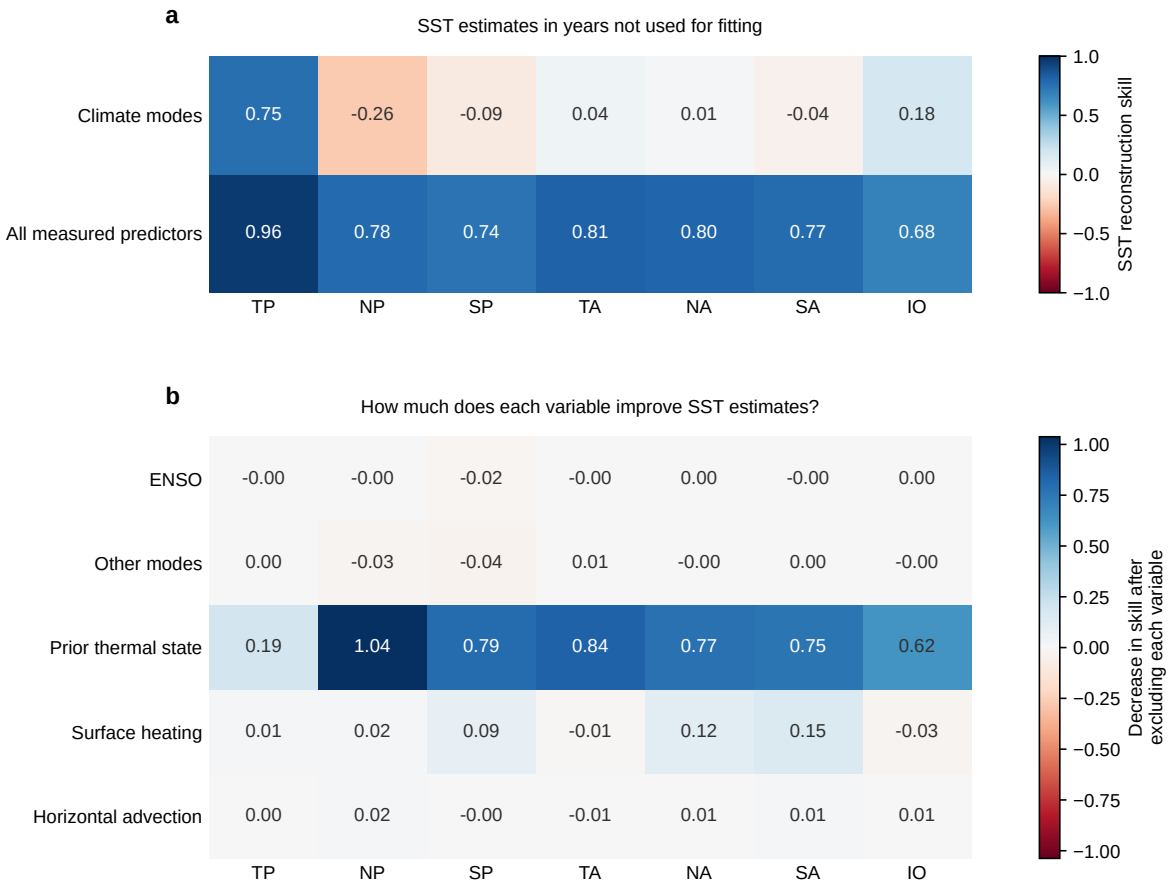
Supplementary Fig. 13 | Testing regional SST estimates in years excluded from fitting



This analysis uses contemporaneous observed climate indices, surface heating and horizontal advection, together with prior SST and mixed-layer temperature; it is diagnostic reconstruction, not an operational forecast. a, Later-period regional SST reconstruction using climate indices alone or all measured predictors. b, Skill decrease after removing each predictor group from the full model; negative values indicate improved reconstruction after removal. Three forward test blocks are 2003–2008, 2009–2014 and 2015–2020. Each model is trained only on earlier months beginning in 1985; seasonal terms, linear trend, scaling and coefficients are fitted within training data. Reconstruction skill is 1 − (model SSE / baseline SSE), where SSE is the sum of squared errors and the baseline is the training-fitted seasonal cycle plus linear trend. Squared errors are summed across test blocks before taking the ratio.

All seven regions and seven predictor configurations are retained. The three successive test blocks are not independent ENSO events. This score is not MHW AUC and does not validate the causal interpretation of the Sankey diagram.